

Отчёт ЛР2 ИГИ
353504 Левшуков Дмитрий Александрович (35350074)

2.

```
D:\Programming\BSUIR\353504_LEVSHUKOV_16\IGI\LR2>docker

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build        Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images       List images
  login        Authenticate to a registry
  logout       Log out from a registry
  search       Search Docker Hub for images
  version      Show the Docker version information
  info         Display system-wide information

Management Commands:
  ai*          Ask Gordon - Docker Agent
  builder      Manage builds
  buildx*      Docker Buildx
  compose*     Docker Compose
  container    Manage containers
  context       Manage contexts
  debug*       Get a shell into any image or container
  desktop*     Docker Desktop commands (Beta)
  dev*         Docker Dev Environments
  extension*   Manages Docker extensions
  feedback*    Provide feedback, right in your terminal!
  image        Manage images
  init*        Creates Docker-related starter files for your project
  manifest     Manage Docker image manifests and manifest lists
  network      Manage networks
  plugin       Manage plugins
  sbom*        View the packaged-based Software Bill Of Materials (SBOM) for an image
  scout*       Docker Scout
  system       Manage Docker
  trust        Manage trust on Docker images
  volume       Manage volumes

Swarm Commands:
  swarm        Manage Swarm

Commands:
  attach       Attach local standard input, output, and error streams to a running container
  commit       Create a new image from a container's changes
```

```
D:\Programming\BSUIR\353504_LEVSHUKOV_16\IGI\LR2>docker run --help
```

Usage: docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

Aliases:

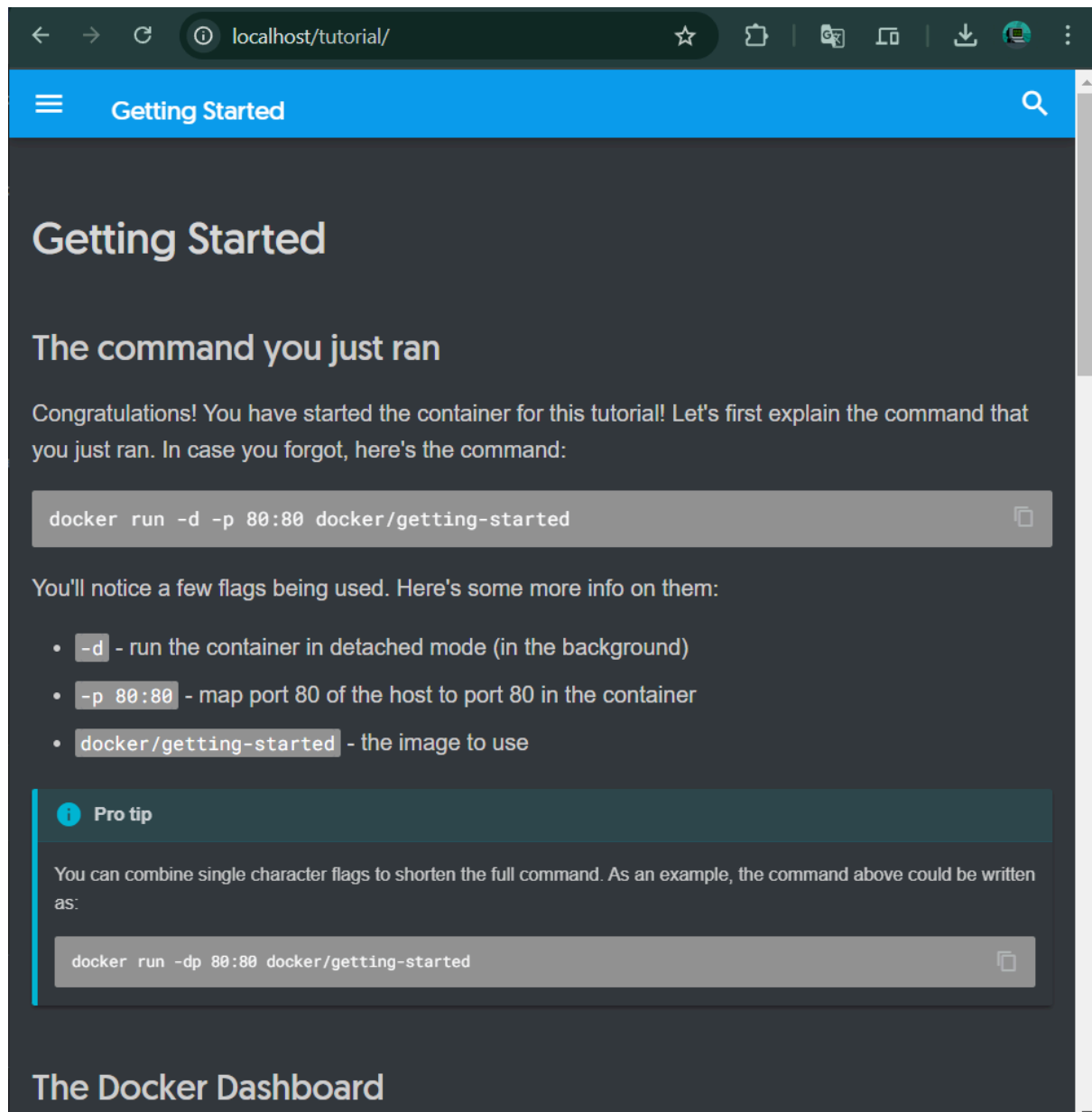
docker container run, docker run

Options:

--add-host list	Add a custom host-to-IP mapping (host:ip)
--annotation map	Add an annotation to the container (passed through to the OCI runtime) (default map[])
-a, --attach list	Attach to STDIN, STDOUT or STDERR
--blkio-weight uint16	Block IO (relative weight), between 10 and 1000, or 0 to disable (default 0)
--blkio-weight-device list	Block IO weight (relative device weight) (default [])
--cap-add list	Add Linux capabilities
--cap-drop list	Drop Linux capabilities
--cgroup-parent string	Optional parent cgroup for the container
--cgroupns string	Cgroup namespace to use (host private) 'host': Run the container in the Docker host's cgroup namespace 'private': Run the container in its own private cgroup namespace '': Use the cgroup namespace as configured by the default-cgroupns-mode option on the daemon (default)
--cidfile string	Write the container ID to the file
--cpu-period int	Limit CPU CFS (Completely Fair Scheduler) period
--cpu-quota int	Limit CPU CFS (Completely Fair Scheduler) quota
--cpu-rt-period int	Limit CPU real-time period in microseconds
--cpu-rt-runtime int	Limit CPU real-time runtime in microseconds
-c, --cpu-shares int	CPU shares (relative weight)
--cpus decimal	Number of CPUs
--cpuset-cpus string	CPUs in which to allow execution (0-3, 0,1)

```
D:\Programming\BSUIR\353504_LEVSHUKOV_16\IGI\LR2>docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
b6334b6ace34: Download complete
33e0cbbb4673: Download complete
9b6f639ec6ea: Download complete
4f7e34c2de10: Download complete
1e35f6679fab: Download complete
f1d1c9928c82: Download complete
c158987b0551: Download complete
ee68d3549ec8: Download complete
cb9626c74200: Download complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
653350a9d5f5c5814473ce486a00b350d0963fdea972dabccd6ec975e44a8b5c

D:\Programming\BSUIR\353504_LEVSHUKOV_16\IGI\LR2>
```



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Getting Started

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
docker run -d -p 80:80 docker/getting-started
```

You'll notice a few flags being used. Here's some more info on them:

- `-d` - run the container in detached mode (in the background)
- `-p 80:80` - map port 80 of the host to port 80 in the container
- `docker/getting-started` - the image to use

Pro tip

You can combine single character flags to shorten the full command. As an example, the command above could be written as:

```
docker run -dp 80:80 docker/getting-started
```

The Docker Dashboard

```

dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker build -t geometric_lib .
[+] Building 4.4s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile              0.0s
=> => transferring dockerfile: 237B                             0.0s
=> [internal] load metadata for docker.io/library/python:latest 1.5s
=> [auth] library/python:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:385ccb8304f633073 0.0s
=> => resolve docker.io/library/python:latest@sha256:385ccb8304f633073 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 414B                                   0.0s
=> CACHED [2/5] RUN apt-get update && apt-get install -y git    0.0s
=> [3/5] RUN git clone https://github.com/smartigaorg/geometric_lib /a 1.5s
=> [4/5] COPY main.py /app                                       0.1s
=> [5/5] WORKDIR /app                                             0.1s
=> exporting to image                                           0.8s
=> => exporting layers                                           0.4s
=> => exporting manifest sha256:a804a3ab1df428885274db719995421d5ee177 0.0s
=> => exporting config sha256:18170c43b8fd72c4538cc4643959dd38fdbadcf8 0.0s
=> => exporting attestation manifest sha256:685ac3699f0aee9f4aee2f7665 0.1s
=> => exporting manifest list sha256:db0dab01cbff02f1903878c6a5fb9d08 0.0s
=> => naming to docker.io/library/geometric_lib:latest          0.0s
=> => unpacking to docker.io/library/geometric_lib:latest       0.2s
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ 

```

```

dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
geometric_lib        latest          db0dab01cbf    43 seconds ago  1.51GB
docker/welcome-to-docker latest          eedaff45e3c7   16 months ago  29.5MB
docker/getting-started latest          d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker run --rm -e SHAPE=circle -e
IZE=3 geometric-lib
Unable to find image 'geometric-lib:latest' locally
docker: Error response from daemon: pull access denied for geometric-lib, repository does
ot exist or may require 'docker login'

Run 'docker run --help' for more information
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker run --rm -e SHAPE=circle -e
IZE=3 geometric_lib
S: 28.274333882308138
P: 18.84955592153876
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ 

```

```

dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
geometric_lib        latest          db0dab01cbf    44 minutes ago  1.51GB
docker/welcome-to-docker latest          eedaff45e3c7   16 months ago  29.5MB
docker/getting-started latest          d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker tag geometric_lib lab
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
geometric_lib        latest          db0dab01cbf    44 minutes ago  1.51GB
lab                  latest          db0dab01cbf    44 minutes ago  1.51GB
docker/welcome-to-docker latest          eedaff45e3c7   16 months ago  29.5MB
docker/getting-started latest          d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker image rm lab
Untagged: lab:latest
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
geometric_lib        latest          db0dab01cbf    45 minutes ago  1.51GB
docker/welcome-to-docker latest          eedaff45e3c7   16 months ago  29.5MB
docker/getting-started latest          d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker tag geometric_lib lab:lab
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
geometric_lib        latest          db0dab01cbf    45 minutes ago  1.51GB
lab                  lab            db0dab01cbf    45 minutes ago  1.51GB
docker/welcome-to-docker latest          eedaff45e3c7   16 months ago  29.5MB
docker/getting-started latest          d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ 

```

```

tag does not exist: dzmitryleushukou/lab:1
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
lab                  1              dbe0dab01cbf   55 minutes ago  1.51GB
docker/welcome-to-docker  latest         eedaff45e3c7   16 months ago  29.5MB
docker/getting-started  latest         d79336f4812b   2 years ago    73.9MB
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker tag lab:1 dzmitryleushukou/lab:1
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker image rm lab:1
Untagged: lab:1
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker push dzmitryleushukou/lab:1
The push refers to repository [docker.io/dzmitryleushukou/lab]
5e9ad5aa09b4: Pushed
78a4768b4d29: Pushed
8031108f3cda: Pushed
21754c21aa78: Pushed
14958205946d: Pushed
1f63342c3384: Pushed
1d281e50d3e4: Pushed
854e2aed8deb: Pushed
155ad54a8b28: Pushed
142e7d1234a5: Pushed
4f4fb700ef54: Pushed
447713e77b4f: Pushed
1: digest: sha256:dbe0dab01cbff02f1903878c6a5fb9d08dbe87e257da84a72d5265a2d420c1f7 size: 856
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$

```

dzmitryleushukou/lab

Last pushed 5 minutes ago

Add a description

Add a category

General
Tags
Image Management BETA
Collaborators
Webhooks
Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
1		Image	less than 1 day	5 minutes

[See all](#)

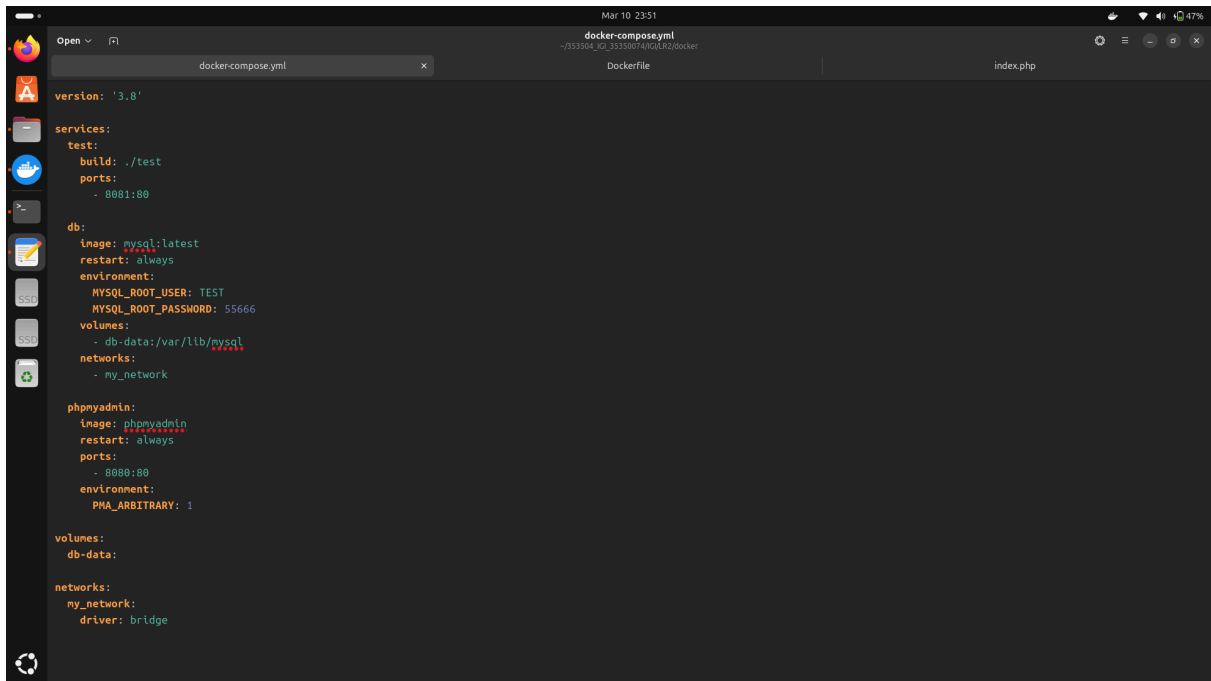
```
Open ▾  docker-compose.yml
~/353504_IGI_35350074/IGI/LR2/docker

services:
  db:
    image: mariadb:10.11
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: notSecureChangeMe

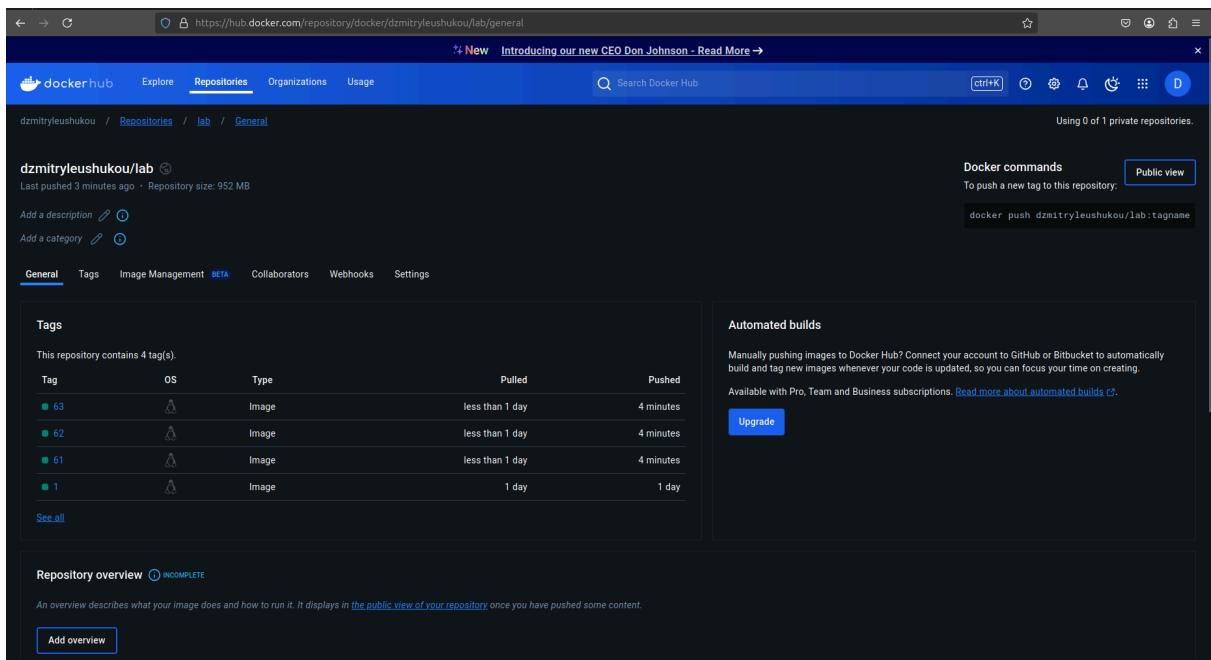
  phpmyadmin:
    image: phpmyadmin
    restart: always
    ports:
      - 8080:80
    environment:
      - PMA_ARBITRARY=1
```

```
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2/docker$ docker compose build
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2/docker$ docker compose up
[+] Running 30/30
 ✓ db Pulled                                73.7s
 ✓ phpmyadmin Pulled                        107.1s

[+] Running 3/3
 ✓ Network docker_default                  Created          0.2s
 ✓ Container docker-phpmyadmin-1          Creat...        2.6s
 ✓ Container docker-db-1                  Created          2.6s
Attaching to db-1, phpmyadmin-1
db-1 | 2025-03-10 18:35:19+00:00 [Note] [Entrypoint]: Entrypoint script
for MariaDB Server 1:10.11.11+maria~ubu2204 started.
phpmyadmin-1 | AH00558: apache2: Could not reliably determine the server's full
y qualified domain name, using 172.19.0.3. Set the 'ServerName' directive global
ly to suppress this message
```

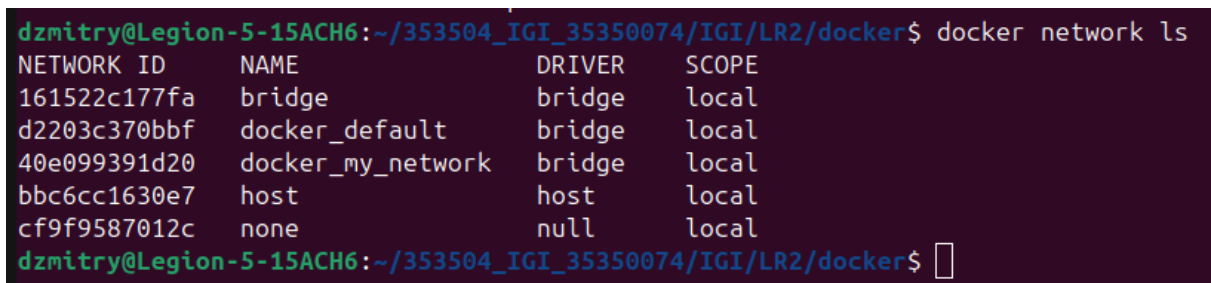


6



7

a




```
Mar 11 00:24
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker
cf9f9587012c none null local
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "161522c177fad96011419fc0af5b11c1cb2b48be68673c62e96f87b041cf23ad",
    "Created": "2025-03-10T21:17:33.684203173Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {}
  }
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect docker_default
[
  {
    "Name": "docker_default",
    "Id": "d2283c370bbf51accf51411eda6c87598ef05ff917efa32862f151a4bc02f4e1",
    "Created": "2025-03-10T20:06:00.570124922Z",
```

```
Mar 11 00:28
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker
"com.docker.network.bridge.enable_icc": "true",
"com.docker.network.bridge.enable_ip_masquerade": "true",
"com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
"com.docker.network.bridge.name": "docker0",
"com.docker.network.driver.mtu": "1500"
},
"Labels": {}
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect docker_default
[
  {
    "Name": "docker_default",
    "Id": "d2283c370bbf51accf51411eda6c87598ef05ff917efa32862f151a4bc02f4e1",
    "Created": "2025-03-10T20:06:00.570124922Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "e7869dcd0af1856c658cbabad8f131128ffb37f03d8454244677d8b7adea0c",
      "com.docker.compose.network": "default",
      "com.docker.compose.project": "docker",
      "com.docker.compose.version": "2.32.4"
    }
  }
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$
```

```
Mar 11 00:28
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker
"Labels": {
  "com.docker.compose.config-hash": "e7869dcd0af1856c658cbabad8f131128ffb37f03d845c4244677d8b7adea0c",
  "com.docker.compose.network": "default",
  "com.docker.compose.project": "docker",
  "com.docker.compose.version": "2.32.4"
}
}
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect docker_my_network
[
  {
    "Name": "docker_my_network",
    "Id": "40e099391d285c169c7df228c02b5cc0992c37e63a03bb30e4d21882427066bc",
    "Created": "2025-03-10T20:06:00.796609135Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "f711e08f95fae0d1d64f49a3c2e3b0ef10e318a1a1f703efb701f36f4eda788",
      "com.docker.compose.network": "my_network",
      "com.docker.compose.project": "docker",
      "com.docker.compose.version": "2.32.4"
    }
  }
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$
```

```
Mar 11 00:28
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect host
[
  {
    "Name": "host",
    "Id": "bbc6cc1630e77fb72bdc4646e5400a66e0d72c036f4b71d0c4d5f1907386547e",
    "Created": "2025-03-10T19:59:29.854308692Z",
    "Scope": "local",
    "Driver": "host",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network inspect none
[
  {
    "Name": "none",
    "Id": "cf9f9587012cbe18564bec219e98de52cf1ea38a249e1633149abe7972845777",
    "Created": "2025-03-10T19:59:29.782082597Z",
    "Scope": "local",
    "Driver": "null",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$
```

b

```
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network create --driver bridge my_bridge
c59528f411e83d3bfdec64b0668ddccfc24bc96de584b150ca569887b3317b5fa
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2/docker$ docker network ls
4NETWORK ID          NAME                DRIVER            SCOPE
286b98bc58e1         bridge             bridge            local
d2203c370bbf         docker_default     bridge            local
40e099391d20         docker_my_network  bridge            local
bbc6cc1630e7         host               host              local
c59528f411e8         my_bridge          bridge            local
cf9f9587012c         none               null              local
```

```

bbc6cc1630e7    host        host        local
c59528f411e8    my_bridge   bridge      local
cf9f9587012c    none        null        local
dzmitry@Legion-5-15ACH6:~$ docker network inspect ^C
dzmitry@Legion-5-15ACH6:~$ docker network inspect c59528f411e8
[
  {
    "Name": "my_bridge",
    "Id": "c59528f411e83d3bfdec64b0668dccfc24bc96de584b150ca569887b3317b5fa",
    "Created": "2025-03-10T21:39:14.828033316Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "7e2e73b293327a1287b92e5b0e49278498acceba4a01066351e9c140871b6dbc": {
        "Name": "quirky_faraday",
        "EndpointID": "d616dfa9f666d1407b2aaf7bd09333e19bc88719f426d1f1bafd59d7be4bca75",
        "MacAddress": "02:42:ac:14:00:02",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]
dzmitry@Legion-5-15ACH6:~$
dzmitry@Legion-5-15ACH6:~$ docker network disconnect c59528f411e8 7e2e73b29332
dzmitry@Legion-5-15ACH6:~$

```

c

```

dzmitry@Legion-5-15ACH6:~$ docker network create --driver bridge brigde2
ddf25edbec58fdd46775aca29f78d4930034dc7dfe4702dbbc8c0018d7568738
dzmitry@Legion-5-15ACH6:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
286b98bc58e1        bridge              bridge              local
ddf25edbec58        brigde2            bridge              local
d2203c370bbf        docker_default      bridge              local
40e099391d20        docker_my_network   bridge              local
bbc6cc1630e7        host                host                local
c59528f411e8        my_bridge           bridge              local
cf9f9587012c        none                null                local
dzmitry@Legion-5-15ACH6:~$
dzmitry@Legion-5-15ACH6:~$ docker run -d --network brigde2 docker/getting-started
b1a88a590c4ce11575e115e84a03dbe9c55fc42d75eb4cf07627e6ccaa9ad1fa
dzmitry@Legion-5-15ACH6:~$ docker run -d --network brigde2 docker/getting-started
db607bdb810513f665bd975e4dff058651380dcf5a588b64f451650dde9574fc
dzmitry@Legion-5-15ACH6:~$ docker run -d --network brigde2 docker/getting-started
1aa2326c8d84d5af22a3af6bb86de5d628abba11a5074fc7f2e17e3f96359b69
dzmitry@Legion-5-15ACH6:~$ docke ps
Command 'docke' not found, did you mean:
  command 'docker' from deb docker.io (24.0.7-0ubuntu4.1)
  command 'docker' from deb podman-docker (4.9.3+ds1-1ubuntu0.2)
Try: sudo apt install <deb name>
dzmitry@Legion-5-15ACH6:~$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
1aa2326c8d84   docker/getting-started "/docker-entrypoint..." 7 seconds ago   Up 7 seconds   80/tcp                             heuristic_pike
db607bdb8105   docker/getting-started "/docker-entrypoint..." 9 seconds ago   Up 8 seconds   80/tcp                             goofy_kowalevski
b1a88a590c4c   docker/getting-started "/docker-entrypoint..." 11 seconds ago   Up 10 seconds   80/tcp                             inspiring_kepler
3bda4b415b47   fe8e5b08cba6        "/docker-entrypoint..." 2 hours ago    Up About a minute 0.0.0.0:8080->80/tcp               docker-phpmyadmin-
c4392443ab19   146682692a3a        "docker-entrypoint.s..." 2 hours ago    Up About a minute 3306/tcp, 33060/tcp               docker-db-1
dzmitry@Legion-5-15ACH6:~$

```

d

```

dzmitry@Legion-5-15ACH6:~$ docker run -it -d --name A1 --network brigde2 docker/getting-started ash
ca31c71fc0dc13193c9bfbee280de3fe26c9f49958f1ee09e2fa935951406ae4
dzmitry@Legion-5-15ACH6:~$ docker run -it -d --name A2 --network brigde2 docker/getting-started ash
94755b0552d359ae10e800ccee91b43f9021c3891d9847940e3f5a708de13601
dzmitry@Legion-5-15ACH6:~$ docker run -it -d --name A3 --network brigde2 docker/getting-started ash
792b7e8d3ccac1050ed8f4365a9c2b5796db415b38618b5ede5c9495d7eb6c94
dzmitry@Legion-5-15ACH6:~$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
792b7e8d3ccac1050ed8f4365a9c2b5796db415b38618b5ede5c9495d7eb6c94	docker/getting-started	"/docker-entrypoint..."	4 seconds ago	Up 4 seconds
94755b0552d359ae10e800ccee91b43f9021c3891d9847940e3f5a708de13601	docker/getting-started	"/docker-entrypoint..."	8 seconds ago	Up 8 seconds
ca31c71fc0dc13193c9bfbee280de3fe26c9f49958f1ee09e2fa935951406ae4	docker/getting-started	"/docker-entrypoint..."	14 seconds ago	Up 13 seconds

```

dzmitry@Legion-5-15ACH6:~$ docker container ls

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
792b7e8d3ccac1050ed8f4365a9c2b5796db415b38618b5ede5c9495d7eb6c94	docker/getting-started	"/docker-entrypoint..."	9 seconds ago	Up 9 seconds
94755b0552d359ae10e800ccee91b43f9021c3891d9847940e3f5a708de13601	docker/getting-started	"/docker-entrypoint..."	13 seconds ago	Up 13 seconds
ca31c71fc0dc13193c9bfbee280de3fe26c9f49958f1ee09e2fa935951406ae4	docker/getting-started	"/docker-entrypoint..."	19 seconds ago	Up 18 seconds

```

dzmitry@Legion-5-15ACH6:~$ docker container attach A2
/ # ping -c 2 A1
PING A1 (172.21.0.2): 56 data bytes
64 bytes from 172.21.0.2: seq=0 ttl=64 time=0.133 ms
64 bytes from 172.21.0.2: seq=1 ttl=64 time=0.164 ms

--- A1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.133/0.148/0.164 ms
/ # ping -c 2 A2
PING A2 (172.21.0.3): 56 data bytes
64 bytes from 172.21.0.3: seq=0 ttl=64 time=0.034 ms
64 bytes from 172.21.0.3: seq=1 ttl=64 time=0.082 ms

--- A2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.034/0.058/0.082 ms
/ # ping -c 2 A3
PING A3 (172.21.0.4): 56 data bytes

```

e

```

dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker swarm init
Swarm initialized: current node (l9nLk9f6jrdh172db2bzq0tL) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-18obk4xpvh27n8gndmwt2dc1cj2k18oBakqlzglnksyjjzxw1p-04t3f4eeke4o4gi9bsq4mmnit 192.168.65.9:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
241a4f9ebd84        bridge              bridge              local
ddf25edbec58        bridge2             bridge              local
d2203c370bbf        docker_default      bridge              local
68d0a99304dc        docker_gwbridge     bridge              local
40e099391d20        docker_my_network   bridge              local
bbc6cc1630e7        host                host                local
kx80rsnw3ztb        ingress             overlay             swarm
c59528f411e8        my_bridge           bridge              local
cf9f9587012c        none                null                local
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network create --driver overlay ov
sfdl381ly6xo53hx8v8vpq1pc
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
241a4f9ebd84        bridge              bridge              local
ddf25edbec58        bridge2             bridge              local
d2203c370bbf        docker_default      bridge              local
68d0a99304dc        docker_gwbridge     bridge              local
40e099391d20        docker_my_network   bridge              local
bbc6cc1630e7        host                host                local
kx80rsnw3ztb        ingress             overlay             swarm
c59528f411e8        my_bridge           bridge              local
cf9f9587012c        none                null                local
sfdl381ly6xo        ov                  overlay             swarm
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$

```

```

dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network inspect ov
[
  {
    "Name": "ov",
    "Id": "sfdl381ly6xo53hx8v8vpq1pc",
    "Created": "2025-03-10T22:50:46.686053143Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4097"
    },
    "Labels": null
  }
]
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$

```

f

```
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network create --driver overlay ov2
is43xamzcg99ixcbfr8u5h8xj
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network ls
NETWORK ID        NAME                DRIVER             SCOPE
241a4f9ebd84      bridge              bridge             local
ddf25edbec58      brigde2            bridge             local
d2203c370bbf      docker_default      bridge             local
68d0a99304dc      docker_gwbridge     bridge             local
40e099391d20      docker_my_network  bridge             local
bbc6cc1630e7      host                host               local
kx80rsnw3ztb      ingress             overlay            swarm
c59528f411e8      my_bridge           bridge             local
cf9f9587012c      none                null               local
sfdl38ily6xo      ov                  overlay            swarm
is43xamzcg99      ov2                 overlay            swarm
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network inspect ov2
[
  {
    "Name": "ov2",
    "Id": "is43xamzcg99ixcbfr8u5h8xj",
    "Created": "2025-03-10T22:51:48.565430918Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.2.0/24",
          "Gateway": "10.0.2.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4098"
    }
  }
]
```

```
dzmitry@Legion-5-15ACH6: ~/353504_IGI_35350074/IGI/LR2
},
  "Internal": false,
  "Attachable": false,
  "Ingress": false,
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Containers": null,
  "Options": {
    "com.docker.network.driver.overlay.vxlanid_list": "4098"
  },
  "Labels": null
}
]
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network --help
Usage:  docker network COMMAND

Manage networks

Commands:
 connect      Connect a container to a network
 create       Create a network
 disconnect   Disconnect a container from a network
 inspect      Display detailed information on one or more networks
 ls           List networks
 prune        Remove all unused networks
 rm           Remove one or more networks

Run 'docker network COMMAND --help' for more information on a command.
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network rm ov2
ov2
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network ls
NETWORK ID        NAME                DRIVER              SCOPE
241a4f9ebd84      bridge              bridge              local
ddf25edbec58      brigde2            bridge              local
d2203c370bbf      docker_default      bridge              local
68d0a99304dc      docker_gwbridge     bridge              local
40e099391d20      docker_my_network   bridge              local
bbc6cc1630e7      host                host                local
kx80rsnw3ztb      ingress             overlay             swarm
c59528f411e8      my_bridge           bridge              local
cf9f9587012c      none                null                local
sfdl38ily6xo      ov                  overlay             swarm
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$
```

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```
sfld38ily6xo  ov                  overlay  swarm
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$ docker network create --driver host
host1
Error response from daemon: only one instance of "host" network is allowed
dzmitry@Legion-5-15ACH6:~/353504_IGI_35350074/IGI/LR2$
```